Housing Discrimination & Algorithmic Bias

Group 1
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Introduction and Historical Context

Historical housing discrimination in the United States has produced long-lasting effects on neighborhood development, property values, and wealth accumulation. Racial covenants were legal clauses placed in property deeds that prevented homes from being sold to non-white residents. These covenants were widely used during the early and mid-twentieth century and contributed to systematic residential segregation. Although these practices were made illegal by the Fair Housing Act of 1968, the spatial and economic patterns created by them remain visible in modern housing data.

This project analyzes how historical housing discrimination may still appear in modern mortgage lending and housing data. The dataset used in the analysis combines several sources, including covenant density by census tract, property value estimates, mortgage lending outcomes, and census demographic indicators. The analysis focuses on Hennepin County, Minnesota. The goal is to identify evidence of data bias, where historical inequality is embedded in modern datasets, and algorithmic bias, where lending systems may reproduce these patterns through the variables they use.

Methodology

The analysis was performed using Python within a Jupyter notebook. The notebook imported pandas for data manipulation, NumPy for numerical operations, matplotlib and seaborn for visualization, and geopandas for spatial data support. The dataset was loaded from a CSV file named mortgage_covenant_data.csv. The census tract column was treated as a string so that geographic codes could be filtered correctly.

To isolate Hennepin County records, the dataset was filtered using the first five digits of the census tract code (27053). This created a subset called df_hennepin containing only records from that county. Exploratory analysis was then performed on this subset. Groupby operations were used to calculate summary statistics such as mean, median, count, and standard deviation for property values and approval outcomes. Several visualizations were created including boxplots, bar charts, heatmaps, and scatterplots to examine relationships between variables.

For property value analysis, the dataset grouped observations by covenant density categories and calculated summary statistics for property_value. Additional analysis converted the covenant_count variable into a Boolean variable called has_covenant indicating whether a tract historically contained at least one covenant. Property value distributions were visualized using boxplots.

For mortgage lending outcomes, approval rates were calculated using the variable was_approved. Approval rates were analyzed overall, by applicant race (derived_race), by covenant density, and by covenant presence. These comparisons helped determine whether historical covenant patterns correspond with modern lending outcomes.

To investigate possible proxy variables, a correlation matrix was computed for several variables including property_value, income, loan_amount, interest_rate, tract_minority_population_percent, tract_to_msa_income_percentage, and covenant_count. A heatmap visualization was generated to highlight strong correlations between variables. Additional grouped summaries compared demographic and economic indicators between tracts with and without historical covenants.

Findings

Property Values and Covenant Density (Data Bias)

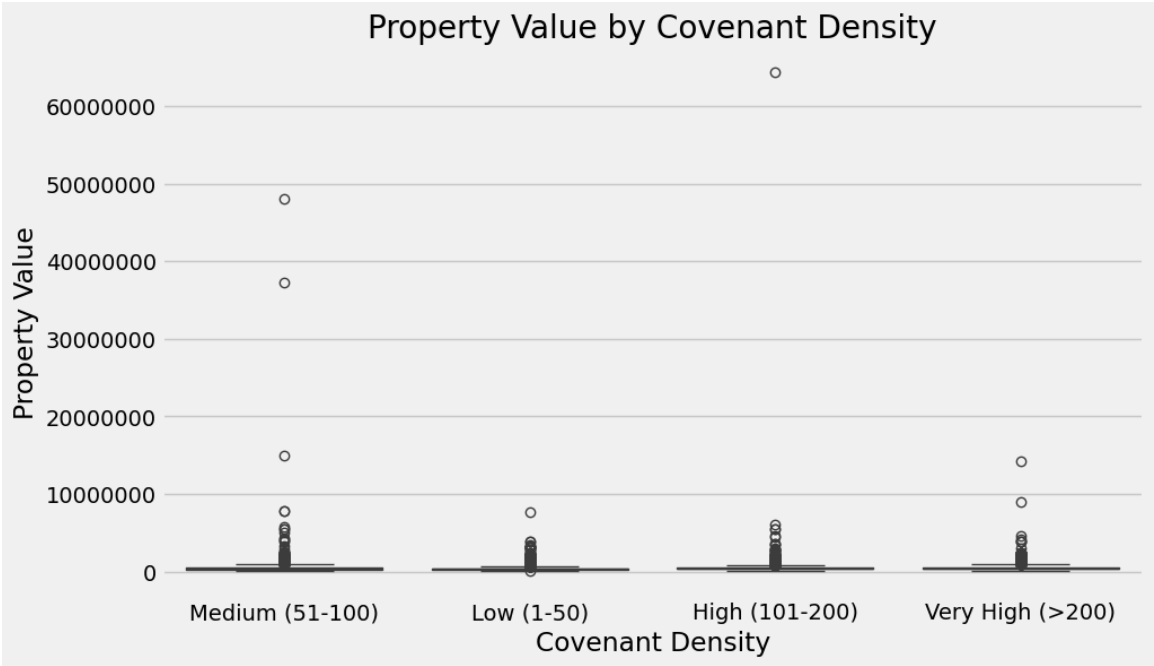

Figure: Property Values and Covenant Density (Data Bias) visualization

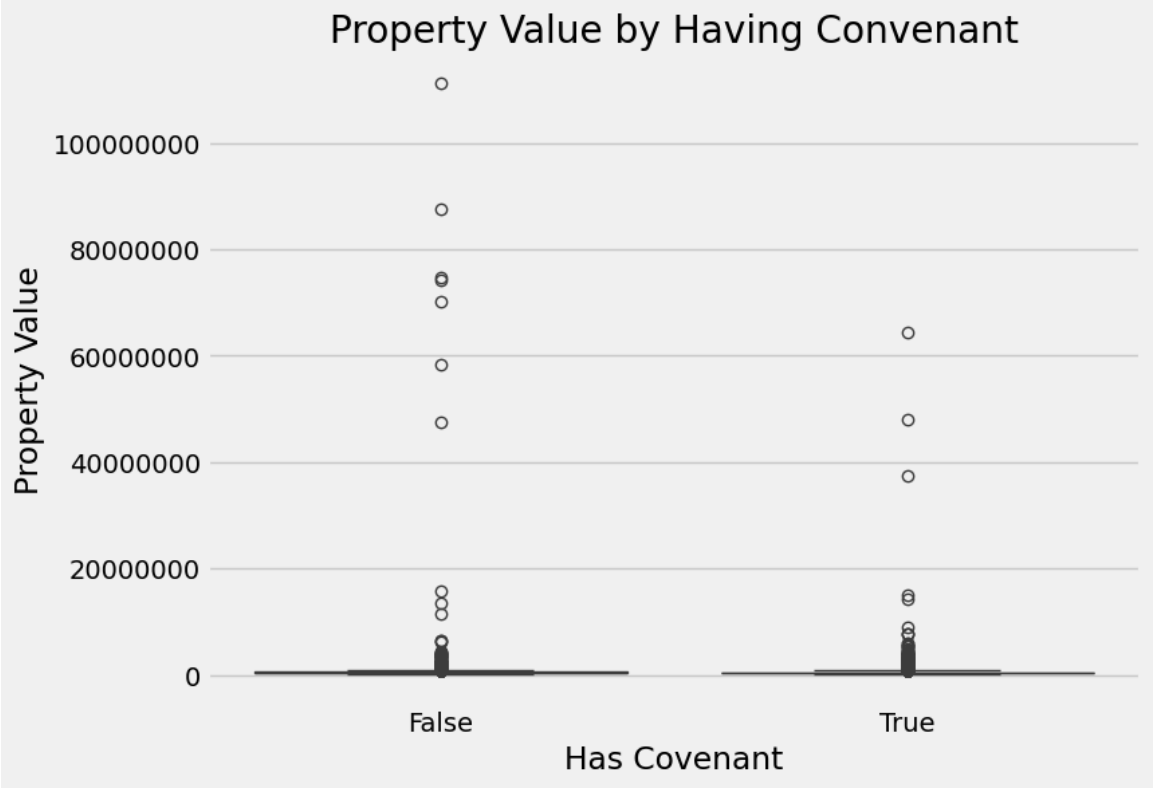

Figure: Property Values and Covenant Density (Data Bias) visualization

Property value patterns were analyzed by grouping census tracts according to covenant density. Summary statistics showed that the 'Low (1–50)' covenant density category had the lowest median property value at approximately $325,000. Other covenant density categories had higher median values, with the 'Very High (>200)' group reaching a median value of about $385,000. However, the mean values for several groups were influenced by extreme high-value outliers.

A boxplot of property_value by covenant_density showed that most home values are clustered in a lower range while a small number of extremely expensive properties extend far above the rest of the distribution. The interquartile ranges for most covenant density categories overlap considerably, indicating that covenant density alone does not fully explain variation in property values.

When the data was simplified into two groups based on covenant presence, areas without historical covenants had a slightly higher median property value of approximately $365,000 compared to $355,000 in covenanted areas. Although the difference is modest, it suggests that historical housing restrictions may still influence economic patterns in residential areas.

Mortgage Approval Patterns (Algorithmic Bias)

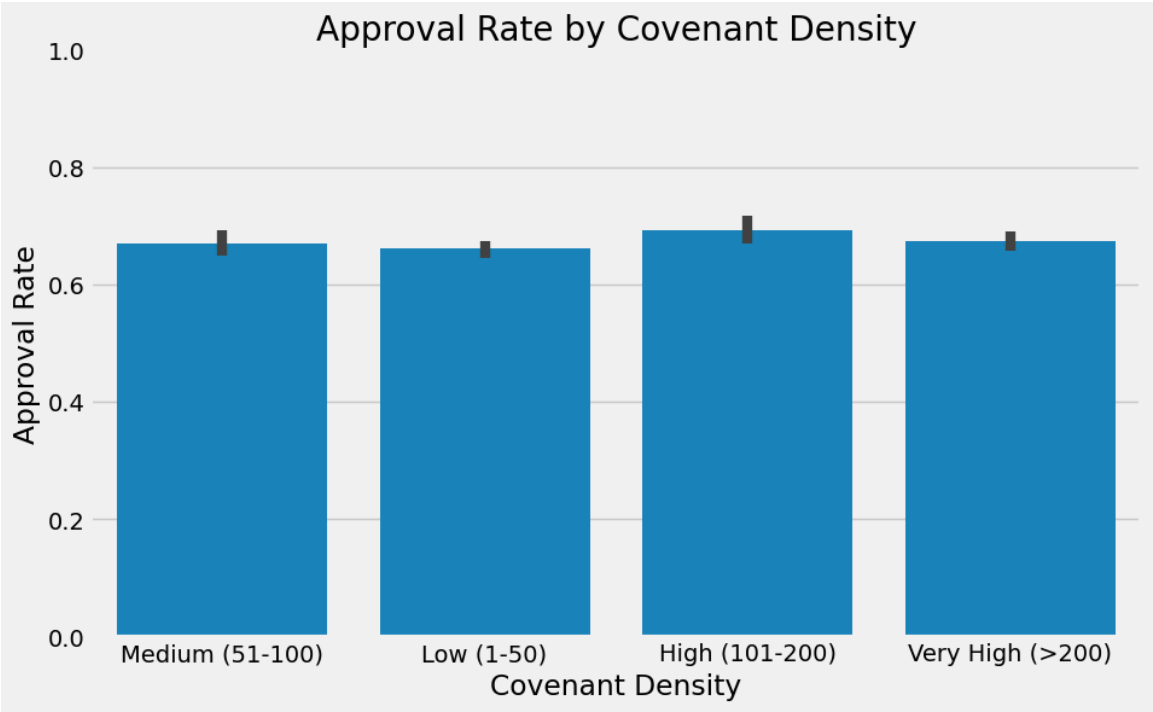

Figure: Mortgage Approval Patterns (Algorithmic Bias) visualization

The overall mortgage approval rate in the Hennepin County dataset was approximately 66 percent. When approval rates were broken down by applicant race, clear disparities appeared. White applicants and joint applications had the highest approval rates, while several other groups including Black or African American and American Indian or Alaska Native applicants had noticeably lower approval rates. The category labeled 'Race Not Available' had the lowest approval rate among all groups.

When approval rates were analyzed by covenant density, the differences were relatively small. The 'High (101–200)' covenant density category had the highest approval rate, while tracts without recorded covenant density had the lowest approval rate. However, the differences between groups were minor and all approval rates were relatively close to the overall average.

A two-group comparison using the has_covenant variable showed that tracts with a historical covenant had an approval rate of about 0.67 while tracts without covenants had an approval rate of about 0.65. This suggests that covenant presence alone does not strongly determine modern mortgage approval outcomes.

Proxy Variables and Demographic Relationships

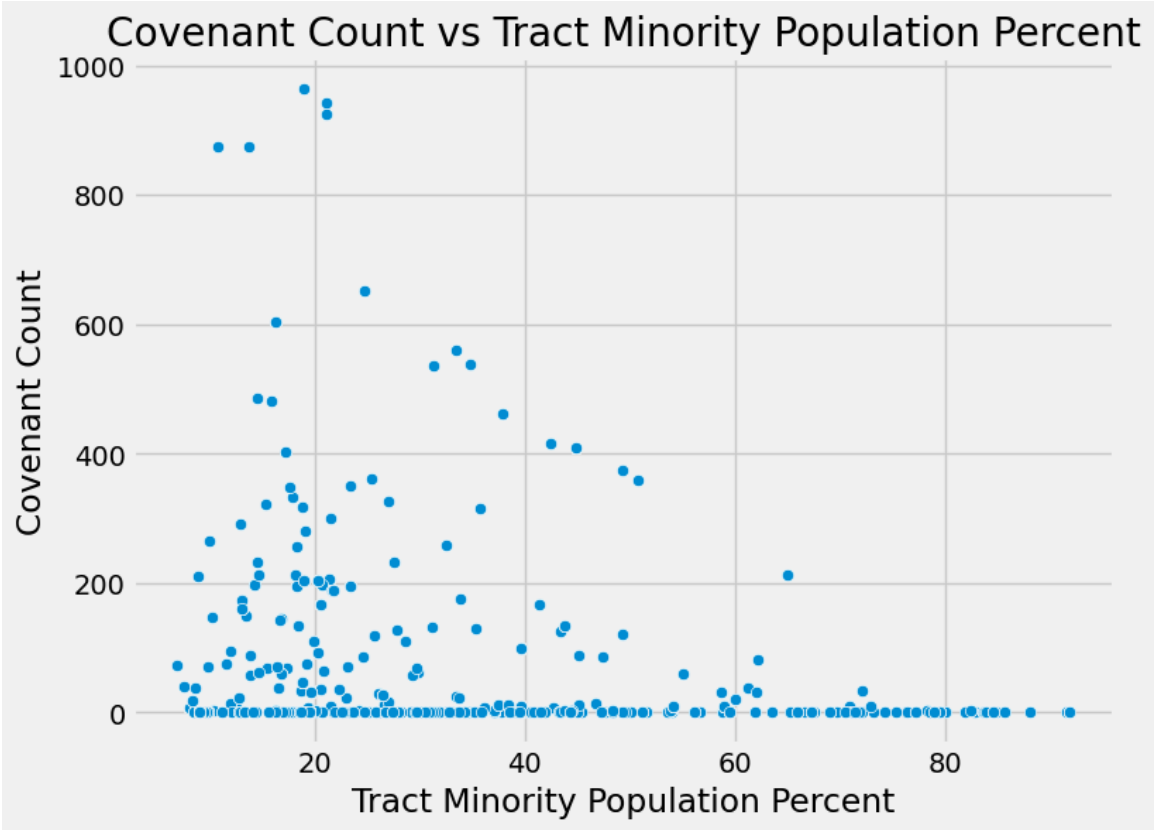

Figure: Proxy Variables and Demographic Relationships visualization

Covenant Status and Racial Disparity

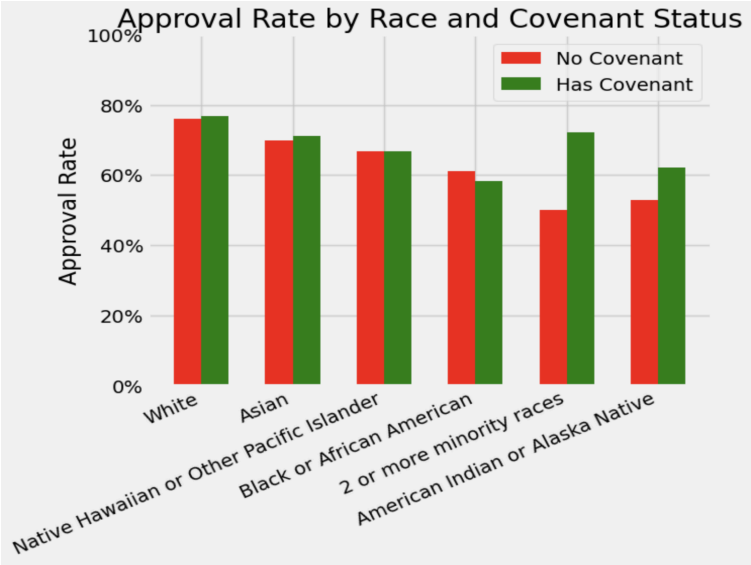

Figure: Covenant Status and Racial Disparity

In the visualization above, properties are separated by whether they sit within a tract that held a racial covenant in the past. A further step is taken to separate these properties by race. As seen in the visualization, for almost all racial groups, approval rates were higher for properties that held a racial covenant. This suggests that the discrimination of the past still has some influence in the algorithms used to determine lending outcomes.

The only race group that showed a lower approval rating in historically covenanted properties as opposed to non covenanted properties is Black or African American. This could be interpreted to show how years of racial discrimination still limit African Americans' access to the neighborhoods they were once banned from purchasing a home in at the algorithmic level.

Conclusion

While historical housing discrimination in the United States has produced long-lasting effects, data analysis can be used to fight that discrimination still seen within today's algorithms. The analysis above showed how discrimination is still embedded in modern data and how that discrimination may still be perpetuated through modern lending algorithms. This modern reflection of historical housing discrimination was examined by looking at Property Values and Covenant Density, Property Values and Covenant Density, Mortgage Approval Patterns, Proxy Variables and Demographic Relationships, and Covenant Status and Racial Disparity.